

Probabilistic Unmixing 0

A Hands-On Tutorial from Known Endmembers to Functional Spectral Priors

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Notebook-driven tutorial for the Probabilistic-Unmixing-0 package

Figures are regenerated from executed notebooks using probunmix v0.1.0.

How to use this tutorial

This tutorial is designed to be read with the notebooks open. The text explains the statistical object being manipulated, the notebook shows the corresponding code, and the figures show an actual execution of that code. The recommended path is sequential: understand the common data object first, then relax spectral knowledge from Level 0 to Level 2, and only then study heterogeneous priors, covariance structure, posterior dependence, and experimental data.

Notebook	Purpose	What you should learn
00_Quick_Start_and_Data	Common data object	What is measured, what is uncertain, and what is synthetic truth.
01_Level_0_Known_Endmembers	Exact spectra	Concentration inference when S is fixed.
02_Level_1_Parametric_Spectral_Uncertainty	Parametric spectra	Joint inference of W and named spectral parameters.
03_Level_2_Functional_Spectral_Uncertainty	Functional spectra	Joint inference using an ensemble of complete spectra.
04_Heterogeneous_Component_Uncertainty	Unequal prior quality	How component-specific W and S uncertainties interact.
05_Uncertainty_Tradeoff_Sweep	Information regime map	How concentration and spectral uncertainty trade off.
06_Level_2_Covariance_and_Correlation	Spectral dependence	How complete-function uncertainty creates energy-energy covariance.
07_Level_2_Functional_Modes_and_Prior_Geometry	Latent geometry	How library -> log-PCA -> physical prior works.
08_Level_2_Posterior_Correlation_and_Propagation	Posterior coupling	How fitting mixtures couples spectra, components, and concentrations.
10_Experimental_Data_Template	Real data	How experimental arrays become the same UnmixingData object.

Learning objective. By the end, you should be able to decide what level of spectral prior your experiment supports, construct a valid UnmixingData object, run the corresponding inference, interpret posterior uncertainties and residuals, and recognize when a covariance or credible band is being asked to represent the wrong physical quantity.

1. The problem: two uncertain factors, one measured mixture

At each measurement location i , an instrument records a spectrum over a spectral coordinate E . In the Stage-1 model, the signal is a linear mixture of K globally shared component spectra with location-dependent fractions, as written in Eq. (1).

$$Y_i(E) = \sum_{k=1}^K W_{ik} S_k(E) + \varepsilon_i(E) \quad (1)$$

The inverse problem is to infer the weights W , the component spectra S , or both. What makes the problem scientifically interesting is that neither block usually starts from nothing. Composition may be partially known from an independent assay, stoichiometry, preparation records, or a phase diagram. Spectra may be partially known from pure references, databases, parametric line-shape models, simulation ensembles, or empirical libraries. The tutorial treats these as distinct information channels.

Stage-1 restriction. The current package assumes linear mixing, one globally shared spectrum per component, exchangeable measurement locations, all components present, no baseline/gain/drift/broadening terms, and—specifically for Probabilistic Unmixing 0—composition = mixture fraction. Spatial coupling, local endmember variability, composition-to-phase maps, solid solutions, nonlinear mixing, unknown component count, and Poisson/instrument layers are later extensions.

1.1 Three uncertainties that must not be conflated

- Measurement noise: variability of a new recorded spectrum around the latent mixture signal. In the tutorial it is represented by `sigma_y` and is heteroscedastic.

- Composition uncertainty: incomplete knowledge of W at each location. In the public tutorial API it is entered as W_sigma in fraction space; the inference modules convert it approximately to additive-log-ratio coordinates. A simplex-native coordinate system is the additive-log-ratio transform in Eq. (2).

$$z_{ij} = \log\left(\frac{W_{ij}}{W_{iK}}\right), \quad j = 1, \dots, K - 1 \quad (2)$$

- Spectral uncertainty: incomplete knowledge of the shared endmember functions. At Level 1 it is induced by uncertain named parameters; at Level 2 it is induced by uncertainty in complete spectral functions.

The Integrated Framework emphasizes a further distinction: epistemic spectral uncertainty is uncertainty about one shared function, whereas intrinsic spectral variability means a different function is realized at each location. The package notebooks model the first. More mixtures can reduce epistemic uncertainty because all mixtures constrain the same global endmembers; they do not average away genuine local variability.

2. The three spectral-knowledge levels

Level	Spectral information supplied	What is inferred	Characteristic uncertainty
0	Exact fixed endmember spectra S_fixed	Local fractions W	Spectral variance is zero by assumption.
1	Parametric model plus θ_mean and θ_sigma	Local W + global spectral parameters	Low-dimensional, physically named spectral uncertainty.
2	Ensemble $S_library$ of complete spectra	Local W + global functional coefficients	Correlated uncertainty over complete spectral functions.

The level is a representation choice, not a strength ordering. A narrow Level 1 prior can be more informative than a broad Level 2 library; conversely, a Level 2 basis that omits a physically real deformation direction can be overconstraining no matter how wide the retained coefficients are. Rank and prior width are therefore different controls.

3. Notebook 00 — build and inspect the common data object

The repository is intentionally organized so that synthetic and experimental data converge to the same UnmixingData object. Everything downstream—validation, fitting, results, and plotting—is shared.

```
import probunmix as pu

data = pu.generate_example(
    n_locations=60,
    n_channels=160,
    n_library=100,
    seed=42,
)

pu.validate_data(data)
pu.print_data_summary(data)
```

In the executed tutorial run, Y had shape (60, 160), W_mean and W_sigma had shape (60, 4), S_fixed had shape (4, 160), θ_mean and θ_sigma had shape (4, 5), and $S_library$ had shape (4, 100, 160). The hidden W_true , S_true , and Y_clean arrays exist only because this is a synthetic benchmark.

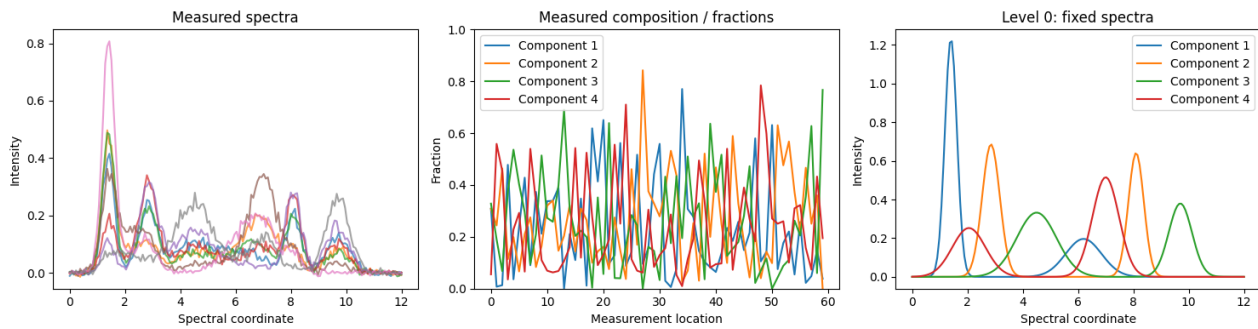


Figure 1. Notebook 00 execution. Representative measured spectra (left), measured composition/fractions across locations (center), and the fixed Level 0 reference spectra (right). The same measured Y and W information will be reused when the spectral prior is changed.

Data contract. Row i of Y , W_mean , W_sigma , and $sigma_y$ must describe the same measurement location. Spectral reference/library arrays must be evaluated on the same spectral coordinate E . This alignment is part of the model, not a plotting convenience.

3.1 What synthetic truth is for

The inference code should never require truth. Truth is present only to calculate recovery error, coverage, and calibration diagnostics. Notebook 10 demonstrates the same workflow after truth has been deliberately removed.

4. Notebook 01 — Level 0: known endmembers

Level 0 is the reference limit. S is fixed and only W is uncertain. In the exact Gaussian theory of the Integrated Framework, this is Bayesian regression on the simplex: the spectral likelihood corrects the external composition estimate in directions that the endmember spectra distinguish well. Non-negativity and closure matter because W lives on a simplex rather than in unconstrained Euclidean space.

```
result0 = pu.fit(
    data,
    level=0,
    n_draws=1000,
    seed=100,
)
```

In the executed Notebook 01 run, reconstruction RMSE was 0.01027, mean posterior concentration SD was 0.00496, and concentration RMSE against synthetic truth was 0.00481. Spectral RMSE and spectral posterior SD were exactly zero because the true endmembers were supplied as exact references.

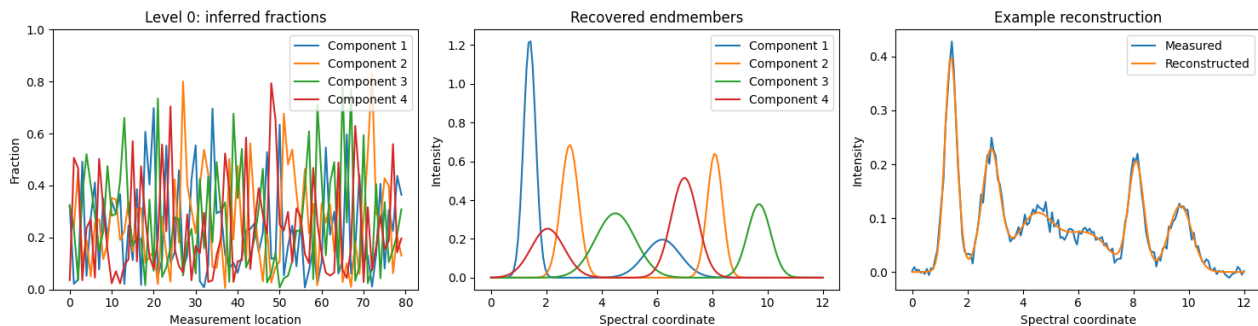


Figure 2. Notebook 01 execution. Posterior concentration trajectories (left), exact recovered endmembers (center), and one measured/reconstructed mixture spectrum (right). Level 0 has no spectral uncertainty by construction.

4.1 Why Level 0 is scientifically useful

- It provides the best-case reference when the spectral model is correct.
- It isolates the effect of concentration uncertainty and measurement noise.

- It exposes spectral-model bias: if an incorrect reference is treated as exact, the inferred fractions can be biased even when spectral reconstruction looks good.
- It supplies a unit test for more complex Level 1 and Level 2 implementations.

Caution. A good reconstruction does not guarantee correct factors. The Integrated Framework derives the exact bias caused by a wrong fixed endmember set; only the component of the spectral error projected into the concentration-sensitive subspace affects W , so residuals can be small while compositions are wrong.

5. Notebook 02 — Level 1: parametric spectral uncertainty

Level 1 replaces exact spectra by a physical line-shape model. The current package uses a two-Gaussian endmember with five parameters per component: relative integrated area r , two peak positions, and two widths. The parameters are global across all locations; the local variables are still the fractions.

```
data = pu.generate_example(spectral_uncertainty=1.0, seed=42)

result1 = pu.fit(
    data,
    level=1,
    n_draws=800,
    seed=101,
    max_nfev=250,
)
```

The executed tutorial run returned nonzero uncertainty in both blocks: mean posterior concentration SD 0.00592 and mean posterior spectral SD 9.28×10^{-4} . On that tutorial-sized dataset, concentration RMSE was 0.00723 and spectral RMSE 0.00265. These values should not be compared numerically with Notebook 01 because the notebook sizes differ; direct cross-level comparison is performed in Notebook 04.

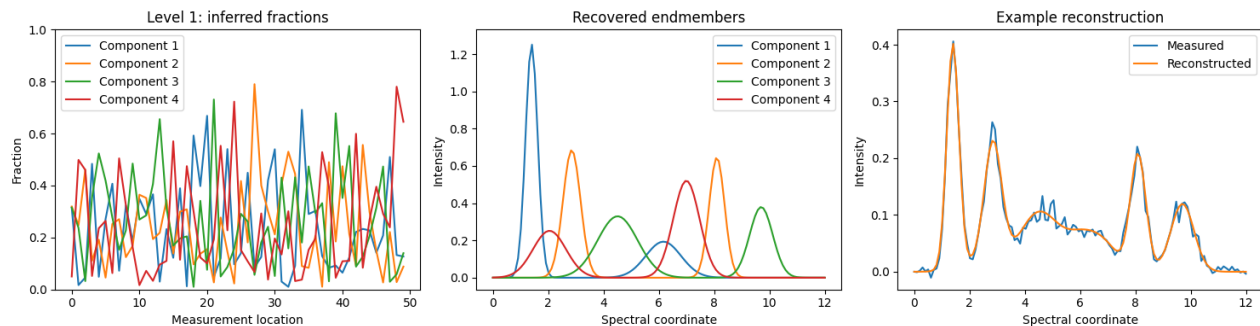


Figure 3. Notebook 02 execution. Level 1 posterior fractions, posterior endmember spectra with uncertainty bands, and a representative spectral reconstruction.

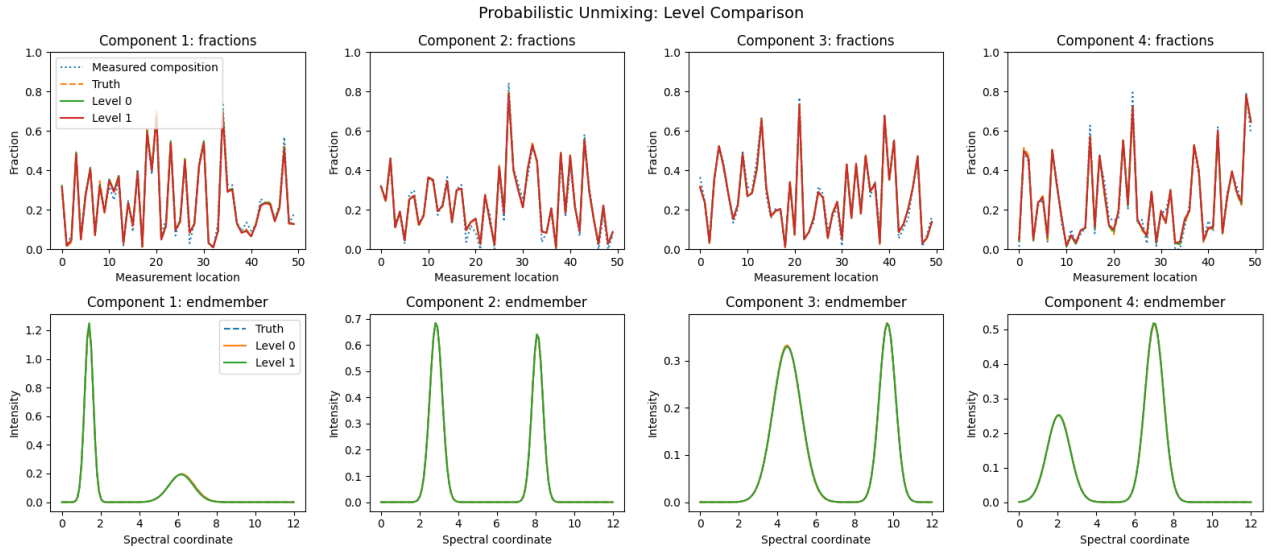


Figure 4. Notebook 02 execution. Level 0 and Level 1 estimates on the same Level 1 tutorial dataset. Relaxing S from exact to parametric adds spectral uncertainty and allows the mixture data to update the shared line-shape parameters.

5.1 What parameter uncertainty means physically

A peak-position uncertainty is not independent intensity noise. A small shift creates a derivative-shaped deformation: one flank rises while the opposite flank falls. Width and area changes create different signed covariance patterns. This is the first reason covariance—not merely pointwise error bars—is central to the framework.

6. Notebook 03 — Level 2: functional spectral uncertainty

Level 2 removes the requirement to name peaks. The input is an ensemble of plausible complete spectra. The current package normalizes the library, moves to log-function space, performs an SVD/PCA for each component, and represents the latent log spectrum by Eq. (3). Independent standard-normal coefficients are then mapped to positive unit-area spectra by Eq. (4).

$$u_k(E) = m_k(E) + B_k(E)a_k \quad (3)$$

$$S_k(E) = \frac{\exp[u_k(E)]}{\int \exp[u_k(E')] dE'} \quad (4)$$

```
result2 = pu.fit(
    data,
    level=2,
    n_draws=800,
    seed=102,
    max_nfev=250,
)
```

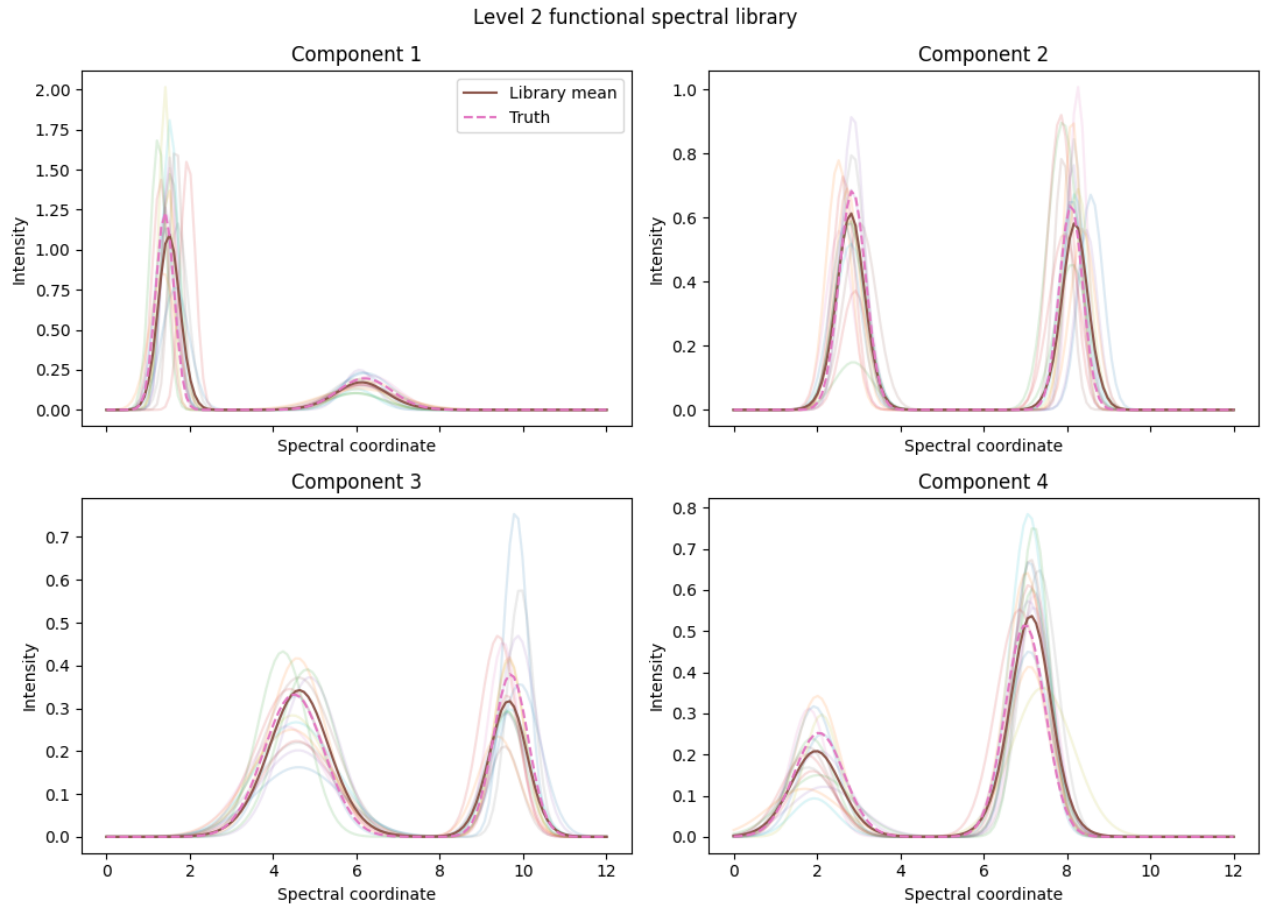


Figure 5. Notebook 03 execution. The Level 2 input library. Faint lines are complete plausible spectral functions; the library mean and hidden truth are overlaid. The uncertainty is coherent across spectral coordinate.

The executed Notebook 03 run retained ranks [12, 12, 10, 9], for 43 global functional parameters. Explained log-library variance was 0.9957, 0.9894, 0.9971, and 0.9950 for the four components. The second component did not reach the nominal 99.5% target under the rank cap, illustrating the source document's warning that rank, variance retention, and physical reconstruction fidelity are separate questions.

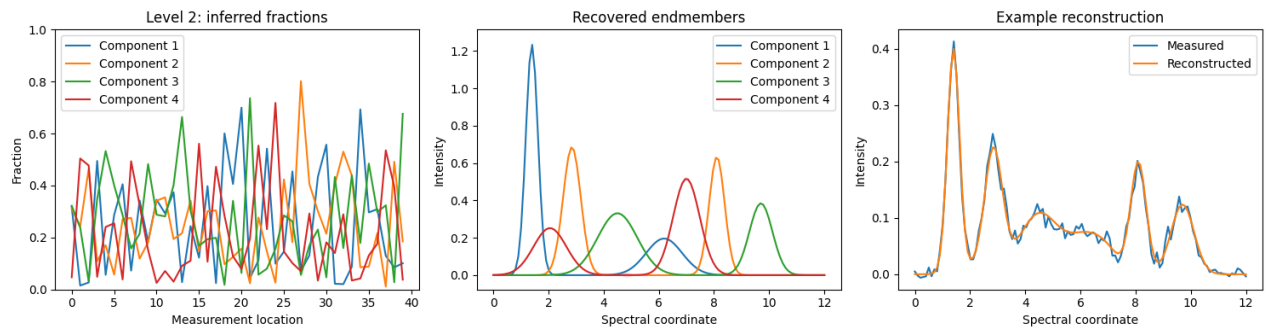


Figure 6. Notebook 03 execution. Posterior fractions, posterior functional endmembers, and a representative mixture reconstruction. The recovered spectra remain positive and numerically unit-area.

6.1 Physical invariants

- Fractions remain nonnegative and sum to one.
- Functional spectra remain positive because the latent representation is exponentiated.
- Every spectrum is normalized to unit integrated area, fixing the scale convention.
- The posterior is conditional on the retained function family; a narrow posterior does not represent uncertainty in omitted functional directions.

Canonical-study difference. The larger Level 2 study in the Integrated Framework built its external library from uncertain two-peak parameters plus seven smooth multiplicative deformations and used HMC for the canonical posterior. The tutorial package generator currently uses parameter perturbations only, and the public fit uses MAP + Gauss-Newton Laplace. Use the notebooks to learn the structure; use stronger inference and richer covariance provenance for scientific certification.

7. Notebook 04 — heterogeneous uncertainty across components

The most important practical generalization is that prior quality is component-specific. Some components may have accurate composition assays but poor spectral references; others may be spectrally distinctive but poorly quantified. The generator therefore accepts vectors rather than only global uncertainty scales.

```
concentration_uncertainty = [0.01, 0.01, 0.10, 0.10]
spectral_uncertainty      = [0.25, 2.00, 0.25, 2.00]

data = pu.generate_example(
    concentration_uncertainty=concentration_uncertainty,
    spectral_uncertainty=spectral_uncertainty,
    seed=42,
)
```

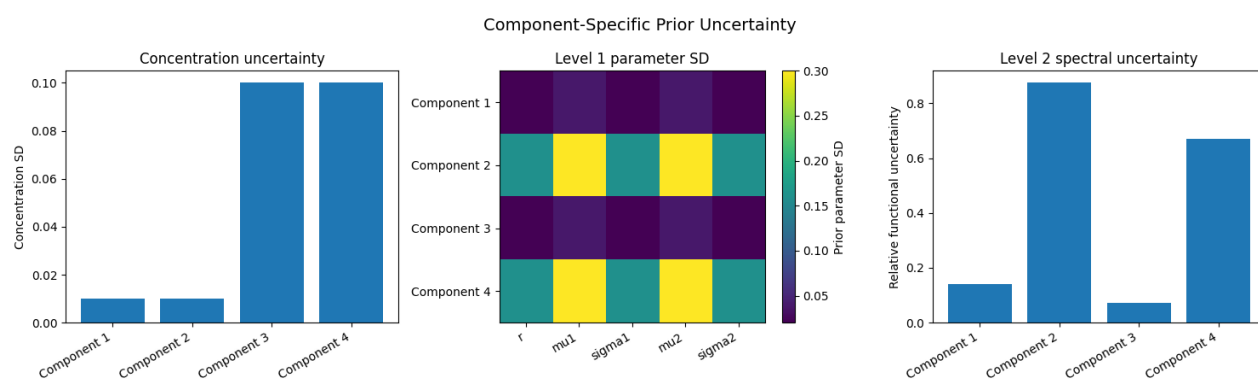


Figure 7. Notebook 04 execution. Component-specific prior uncertainty. Components 1–2 have tight composition information; 3–4 are broad. Components 1 and 3 have narrow spectral priors; 2 and 4 are broad.

The four regimes are deliberately arranged as good/good, good/poor, poor/good, and poor/poor. This makes Components 2 and 3 a controlled contrast: one is composition-anchored and the other spectrum-anchored.

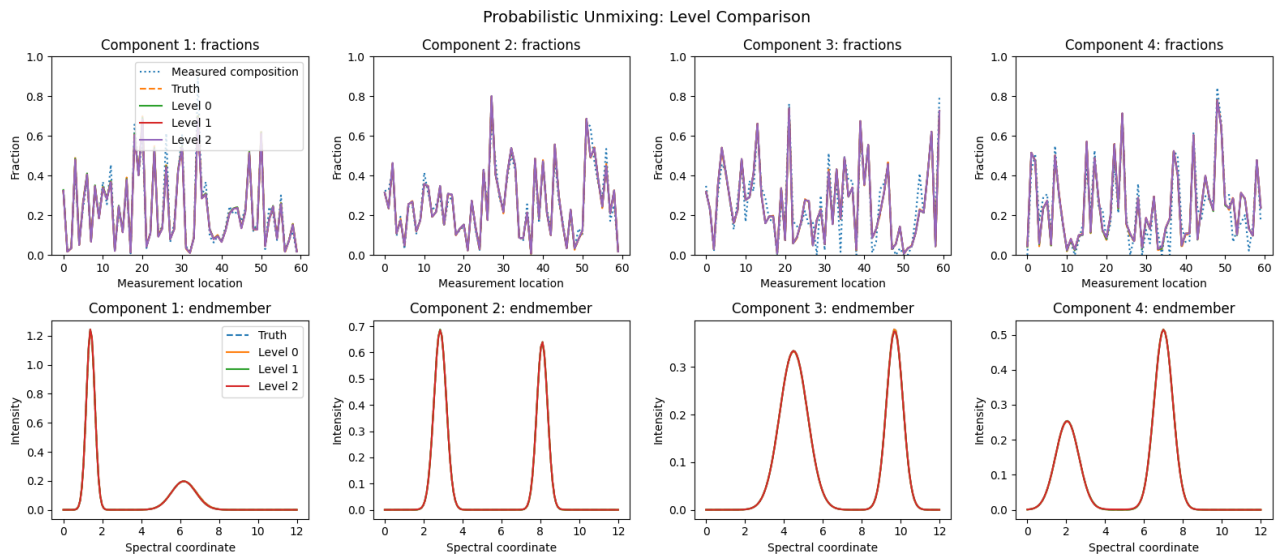


Figure 8. Notebook 04 execution. Direct Level 0/1/2 comparison on one common heterogeneous dataset. The upper row compares component fractions; the lower row compares endmember recovery.

Comp.	Regime	Input W SD	L1 W RMSE	L1 S RMSE	L2 W RMSE	L2 S RMSE	L2 S SD
1	W good / S good	0.01	0.00693	0.00322	0.00701	0.00320	0.00096
2	W good / S poor	0.01	0.00449	0.00112	0.00465	0.00137	0.00101
3	W poor / S good	0.10	0.00544	0.00116	0.00540	0.00115	0.00068
4	W poor / S poor	0.10	0.00636	0.00054	0.00619	0.00096	0.00162

These numbers are single-realization tutorial diagnostics. They demonstrate that posterior uncertainty need not mirror the input ordering exactly: the mixture likelihood transfers information between blocks. The scientifically relevant question is not “which prior was widest?” but “which directions remain weakly identified after all data are combined?”

8. Notebook 05 — map the concentration–spectral trade-off

A single good/poor label hides the transition between regimes. Notebook 05 varies one component's concentration uncertainty and spectral uncertainty while holding the underlying physical experiment fixed. This produces an information-regime map rather than a collection of unrelated random datasets.

Matched physical experiment. At a fixed random seed, S_{true} , W_{true} , Y_{clean} , Y , and σ_y remain identical when uncertainty settings are changed. Only W_{sigma} , θ_{sigma} , and S_{library} change. This is essential for causal interpretation of a sensitivity sweep.

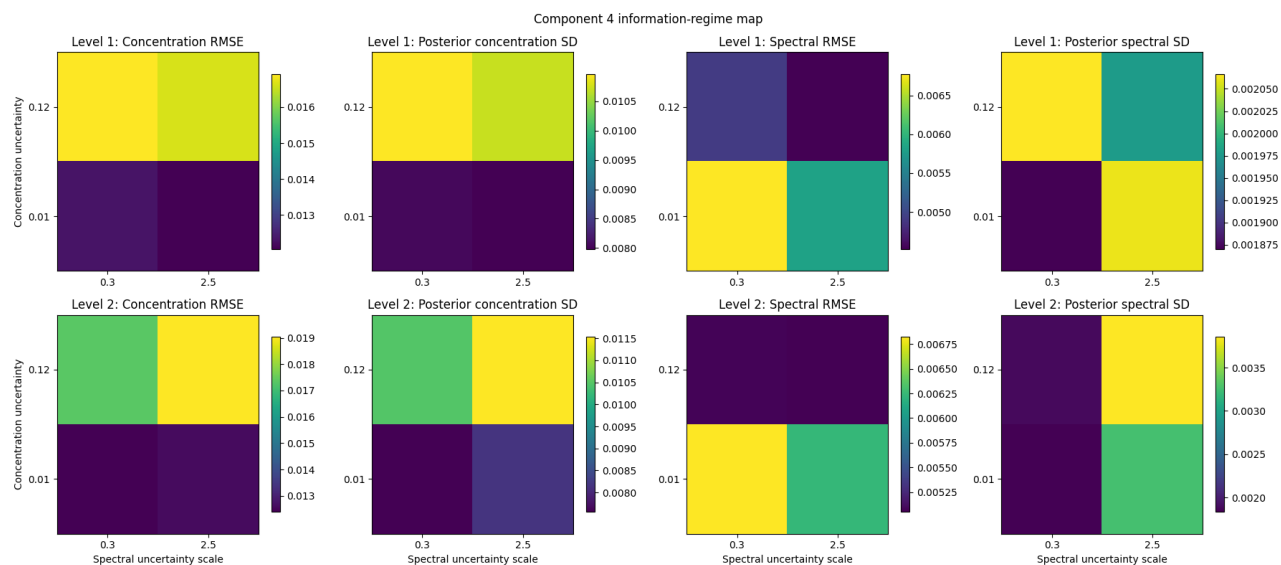


Figure 9. Notebook 05 reduced execution used for this tutorial document. Heat maps show concentration and spectral recovery/uncertainty for Level 1 (top) and Level 2 (bottom) as Component 4 prior quality is changed. The distributed notebook defaults to a denser grid; this figure was executed on a reduced 2×2 grid to keep document-generation time bounded.

Interpretation is geometric. A vertical trend means concentration uncertainty dominates; a horizontal trend means spectral uncertainty dominates. Curved or diagonal contours indicate a genuine trade-off between the two information channels. The Integrated Framework emphasizes that prior width is only one axis of prior knowledge: coverage, precision, and bias should ultimately be varied independently.

9. Notebook 06 — Level 2 covariance and correlation

A pointwise uncertainty band answers a marginal question at each E. It does not describe which intensities can vary together in a physically admissible complete spectrum. Notebook 06 therefore works directly with the covariance of physical spectral draws.

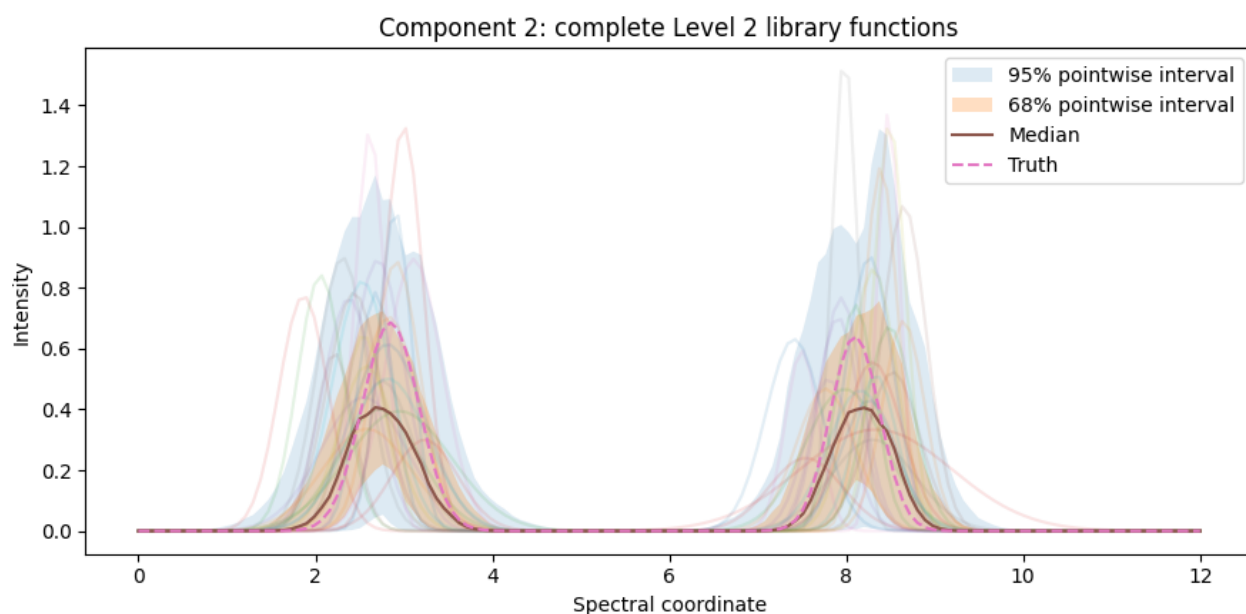


Figure 10. Notebook 06 execution. Component 2 complete library functions and pointwise intervals. Each faint curve is a complete correlated possibility; the shaded bands are marginal summaries.

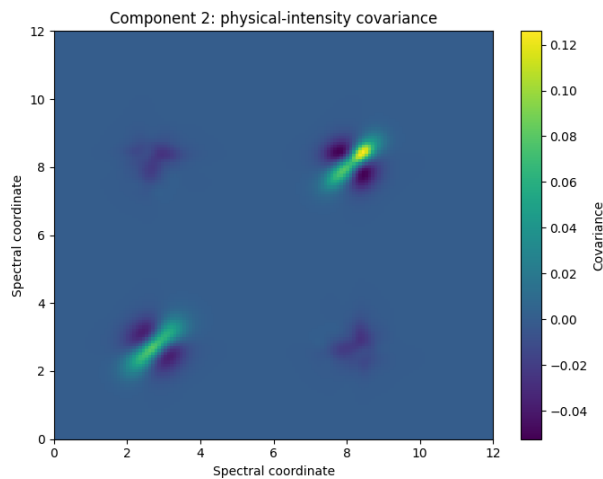


Figure 11. Notebook 06 execution. Physical-intensity covariance of Component 2. Off-diagonal positive and negative regions encode coordinated deformations across spectral coordinate.

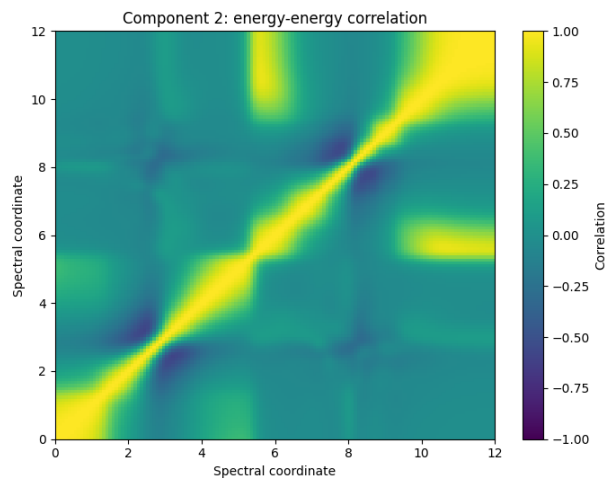


Figure 12. Notebook 06 execution. Energy-energy correlation of Component 2. Correlation removes the marginal variance scale and makes coordinated structure visible, including anticorrelation.

In this execution, the Component 2 correlation ranged from about -0.585 to 1.0, while its maximum pointwise SD was 0.355. The relative functional uncertainty was 0.821, substantially larger than the other three components in this example.

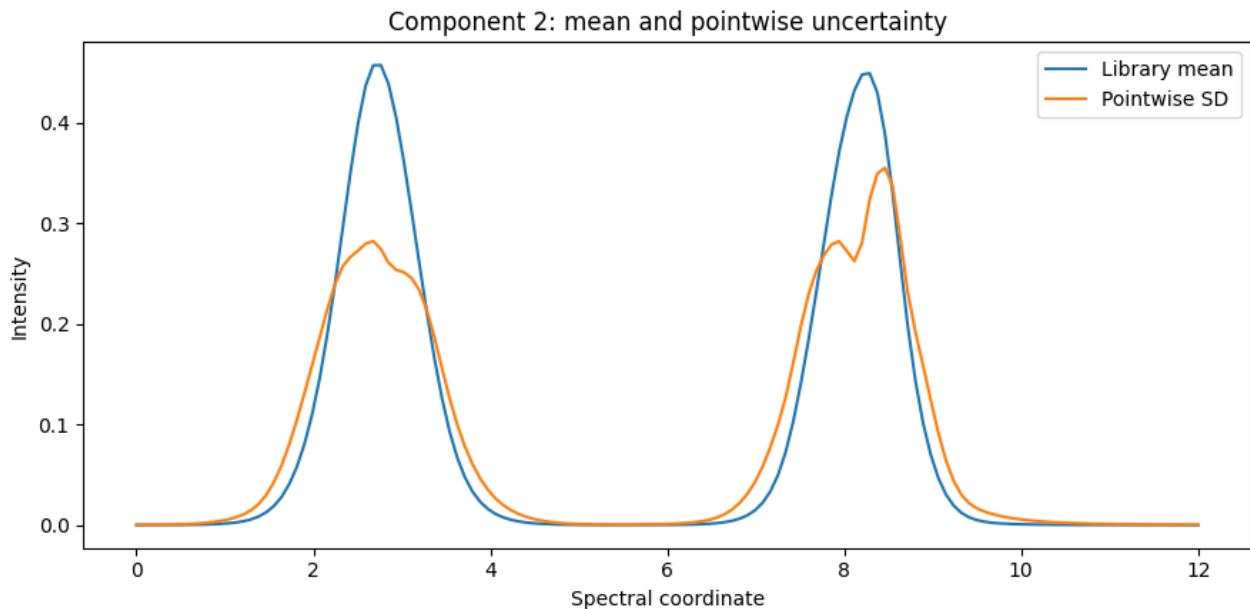


Figure 13. Notebook 06 execution. Library mean and pointwise SD. Strong correlation in a weak tail should not be interpreted as large absolute uncertainty; covariance, correlation, and SD must be read together.

9.1 The unit-area null direction

Because every physical spectral draw integrates to one, fluctuations cannot change total area. If q is the vector of trapezoidal integration weights and K_S is the physical spectral covariance, the exact covariance constraint is Eq. (5). The Notebook 06 execution gave $\max |K_S q| = 1.82 \times 10^{-17}$. This is a useful admissibility test for any supplied physical spectral covariance.

$$K_S q = 0 \quad (5)$$

10. Notebook 07 — functional modes and prior geometry

The Level 2 model contains three different objects that should not be conflated: the empirical library, the low-rank Gaussian distribution in latent log-function space, and the induced nonlinear distribution of positive unit-area physical spectra.

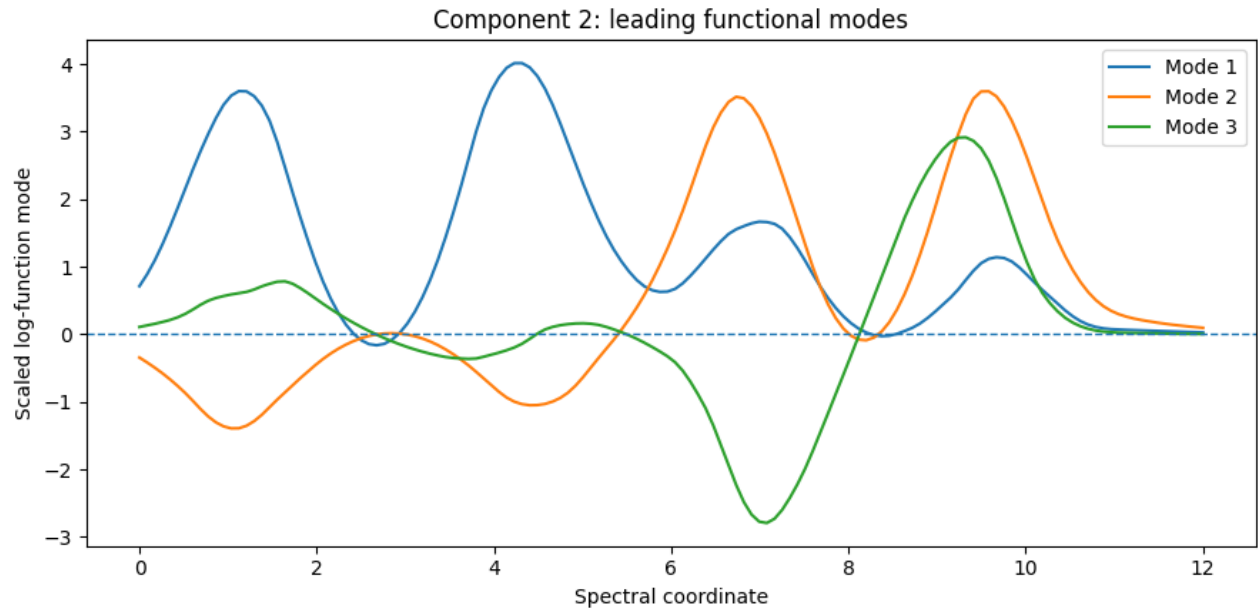


Figure 14. Notebook 07 execution. Leading scaled log-functional modes for Component 2. Modes can be positive or negative because they are directions in latent log space, not spectra.

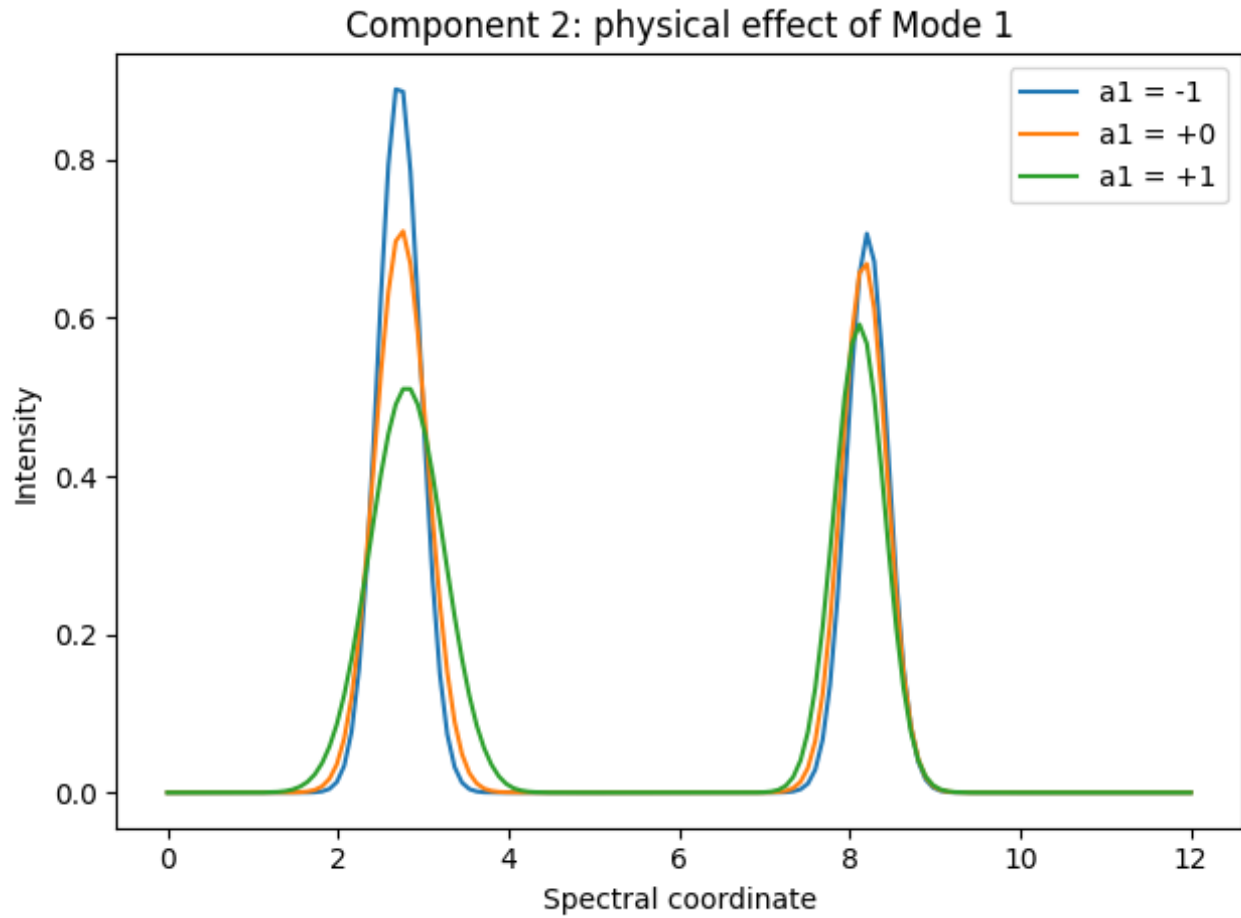


Figure 15. Notebook 07 execution. Physical effect of setting the first functional coefficient to -1, 0, or +1 while all other coefficients remain zero. The nonlinear map converts a latent direction into a complete positive normalized spectrum.

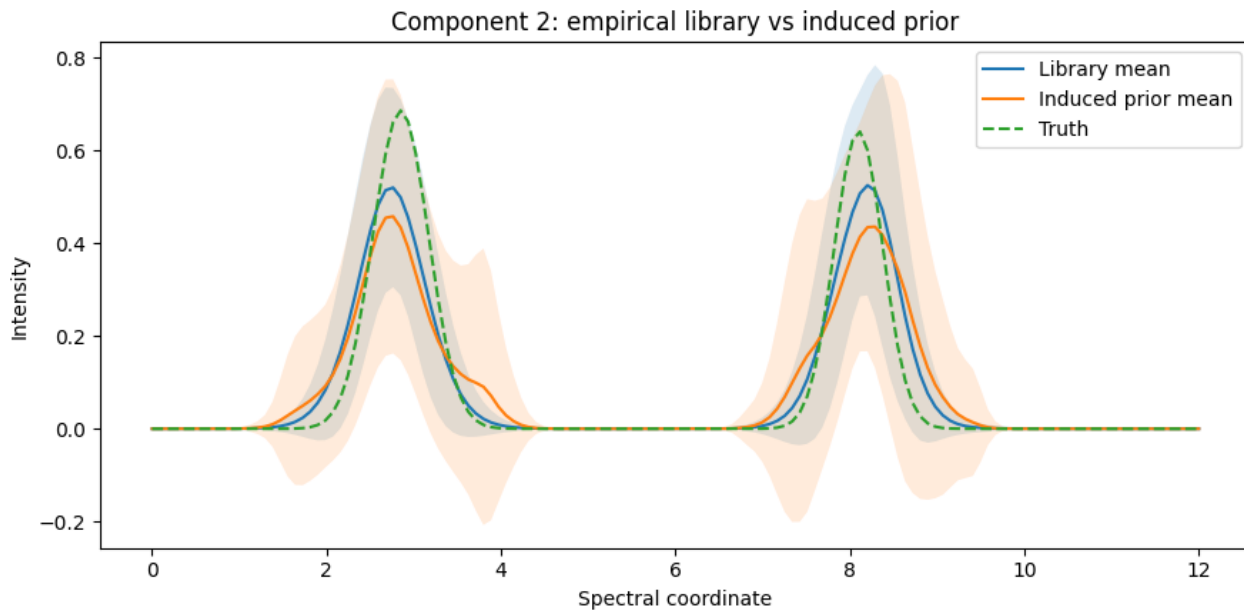


Figure 16. Notebook 07 execution. Empirical library versus induced log-PCA prior for Component 2. Gaussianization, rank truncation, exponentiation, and normalization can change both the mean and the uncertainty.

In this run, Component 2 required 19 modes to reach the nominal variance target but only 12 were retained under the cap, leaving 98.1% explained log-space variance. This is exactly the situation where “variance explained” should not be treated as synonymous with “physical fidelity.”

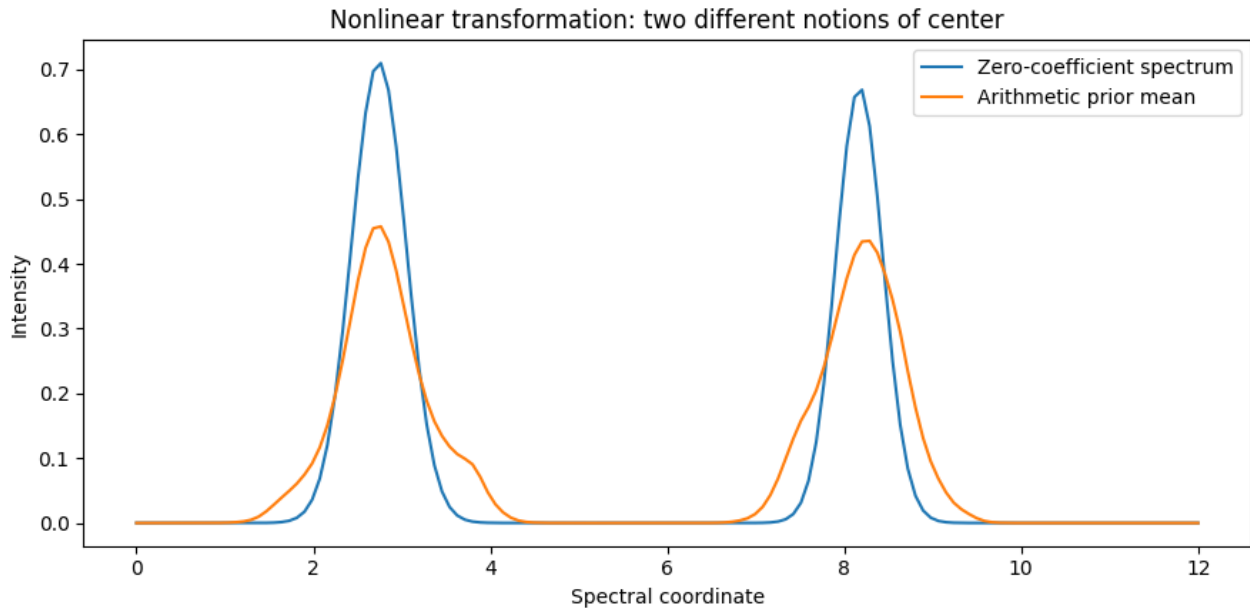


Figure 17. Notebook 07 execution. The zero-coefficient physical spectrum differs from the arithmetic mean of physical prior draws because exponentiation and normalization are nonlinear operations.

The RMS difference between those two notions of center was 0.0741 in the executed example. This is why widening a latent prior can move not only the variance but also the physical prior mean.

11. Notebook 08 — posterior correlation and uncertainty propagation

Before seeing mixture data, the retained Level 2 functional coefficients are independent standard-normal coordinates. After conditioning, they need not remain independent. Spectral modes can compensate each other, different endmembers can compensate each other, and spectral changes can be offset by concentration changes.

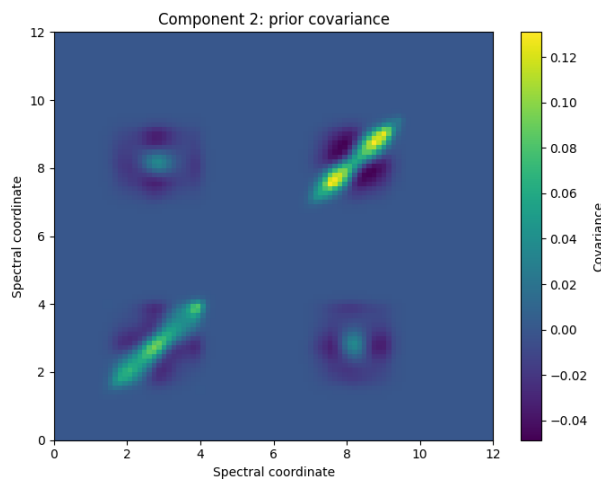


Figure 18. Notebook 08 execution. Prior physical covariance of Component 2 reconstructed from the retained functional prior.

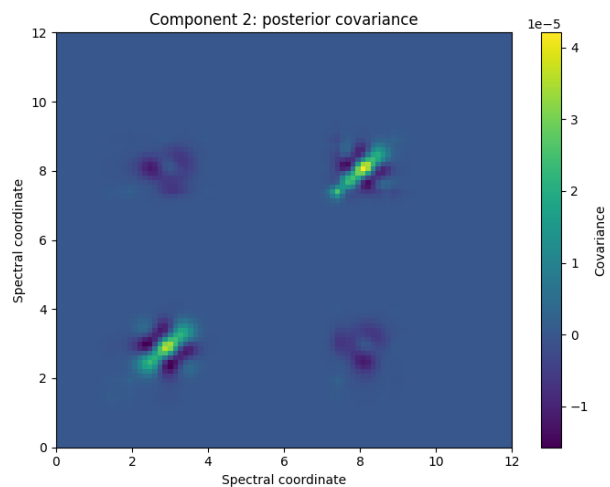


Figure 19. Notebook 08 execution. Posterior physical covariance of the same component after fitting mixture spectra. The absolute scale contracts dramatically while structured dependence remains.

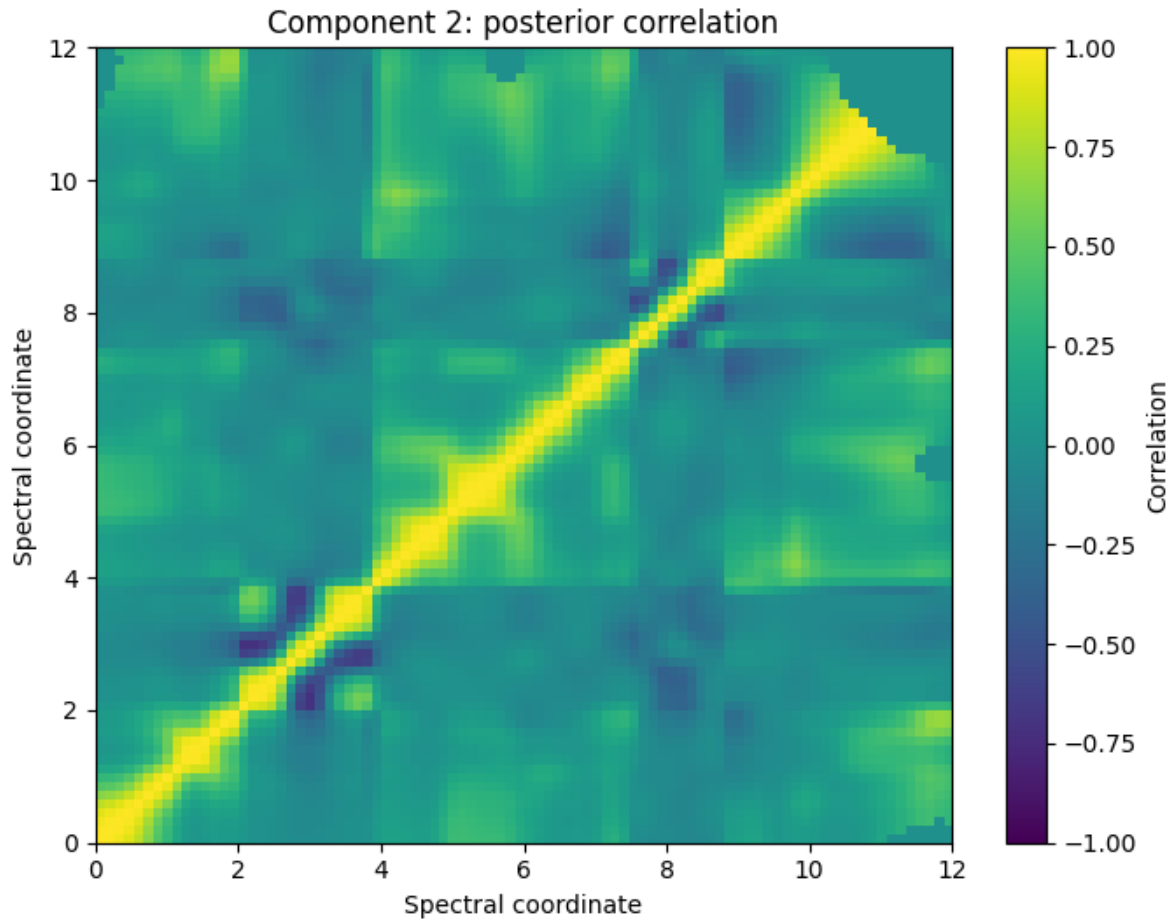


Figure 20. Notebook 08 execution. Posterior energy–energy correlation. Correlation can remain strong even after covariance magnitude has collapsed.

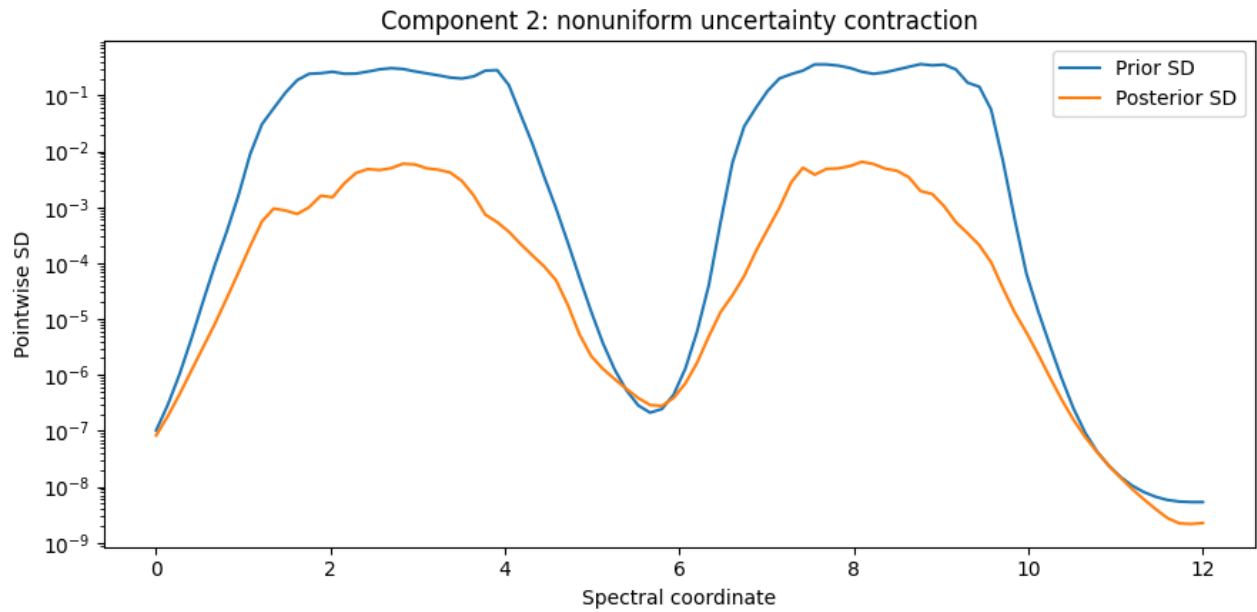


Figure 21. Notebook 08 execution. Prior versus posterior pointwise SD on a logarithmic scale. Uncertainty contraction is strongly energy dependent.

The joint-draw reconstruction agreed with the compact UnmixingResult spectral mean to an RMS of 3.8×10^{-8} . Mean pointwise SD for the selected component contracted from 0.115 in the induced prior to 0.00134 in the posterior—about two orders of magnitude—yet the correlation pattern remained structured.

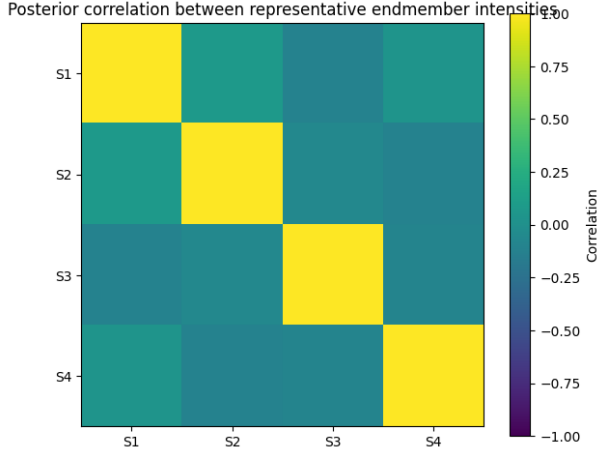


Figure 22. Notebook 08 execution. Posterior correlation between representative endmember intensities. Off-diagonal terms arise because mixture fitting couples components even though their prior coefficients were independent.

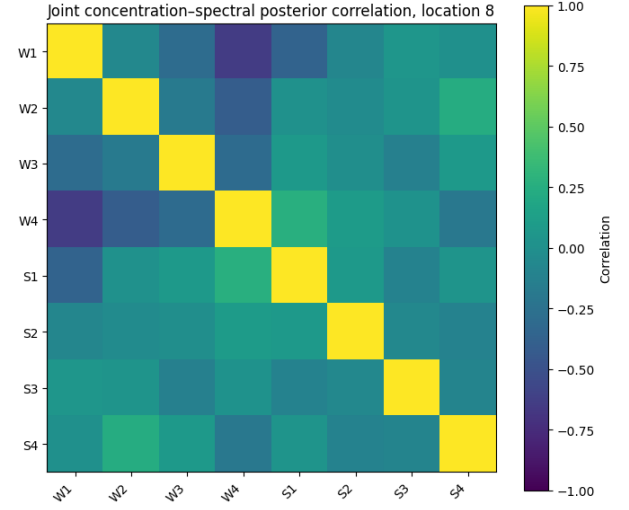


Figure 23. Notebook 08 execution. Joint posterior correlation between four local fractions W1-W4 and representative global spectral intensities S1-S4 at a mixed location. This dependence is destroyed by sampling marginal posteriors independently.

For one location, the fluctuation of the latent mixture around its posterior mean can be decomposed as Eq. (6). Here T_W is the concentration-only term, T_S is the endmember-only term, and T_I is the bilinear interaction term.

$$\Delta y = T_W + T_S + T_I \quad (6)$$

Because these three terms are generally correlated after inference, the total variance contains signed cross-covariances, as summarized in Eq. (7). Here V_W , V_S , and V_I are the variances of the three terms in Eq. (6), while C_Σ is the sum of their three pairwise cross-covariances.

$$V_y = V_W + V_S + V_I + 2C_\Sigma \quad (7)$$

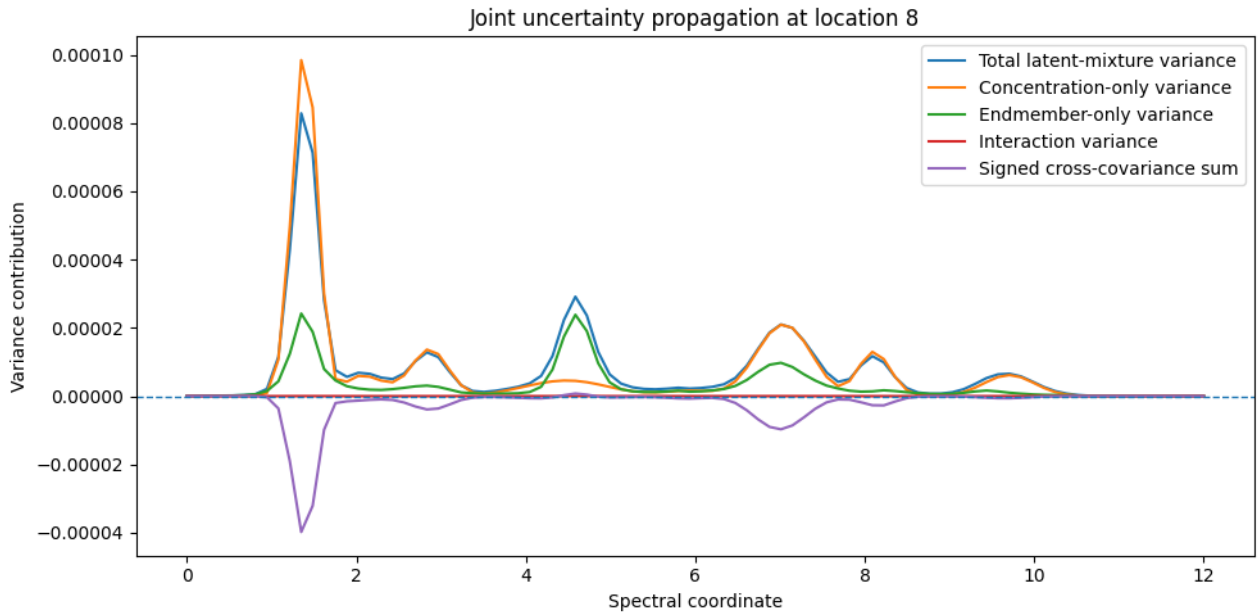


Figure 24. Notebook 08 execution. Exact posterior uncertainty propagation at one location. Concentration-only, endmember-only, bilinear interaction, and signed cross-covariance contributions reconstruct the total latent-mixture variance.

The maximum residual of the variance decomposition in this execution was 1.36×10^{-19} . The numerical result verifies Eq. (7). The key lesson is conceptual: after inference, predictive variance is not generally the sum of independent nonnegative “composition” and “spectral” curves. Paired joint posterior draws are the safest propagation mechanism.

12. Notebook 10 — use the same package on experimental data

Experimental use changes only the entrance to the pipeline. Instead of `generate_example()`, construct `UnmixingData` from measured arrays and then attach the spectral knowledge that is justified by the experiment.

```
data = pu.from_arrays(
    E=E,
    Y=Y,
    W_mean=W_mean,
    W_sigma=W_sigma,
    sigma_y=sigma_y,
)

# choose one spectral-knowledge representation
data.set_level0(S_reference)
# or
data.set_level1(model="two_gaussian", theta_mean=theta_mean, theta_sigma=theta_sigma)
# or
data.set_level2(S_library)

result = pu.fit(data, level=2)
```

Notebook 10 deliberately reconstructs an experimental-style object without `W_true`, `S_true`, or `Y_clean`. The executed summary correctly reports “Synthetic truth: NO”; truth-based concentration and spectral RMSE keys are absent from the result metrics.

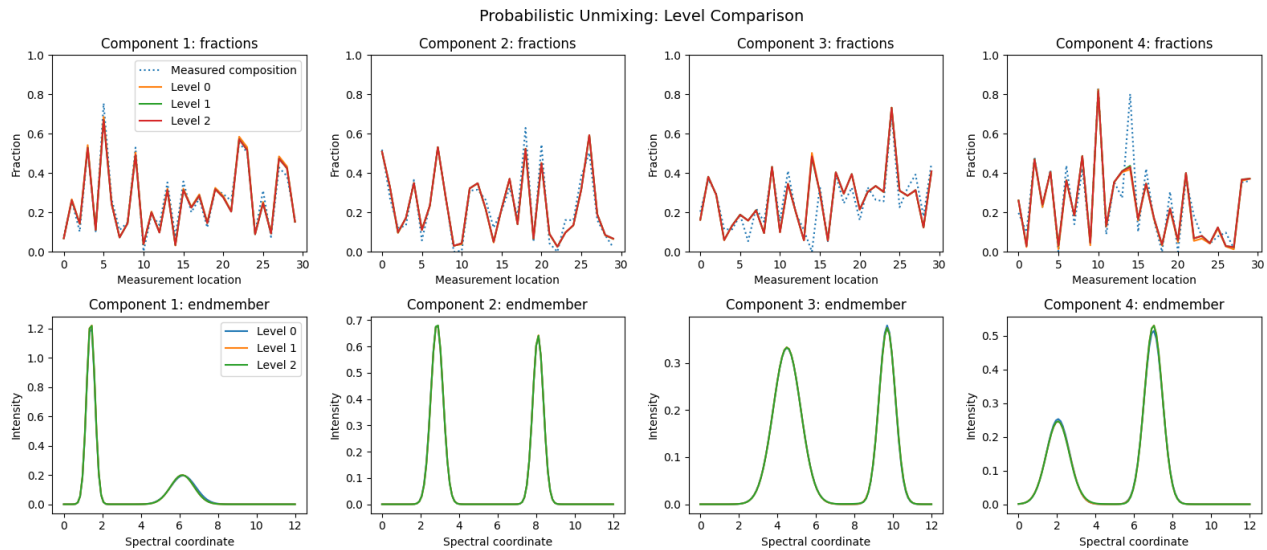


Figure 25. Notebook 10 execution. Cross-level comparison after the same arrays are reloaded as experimental data with no hidden truth attached. The code path is identical to the synthetic case after `UnmixingData` is constructed.

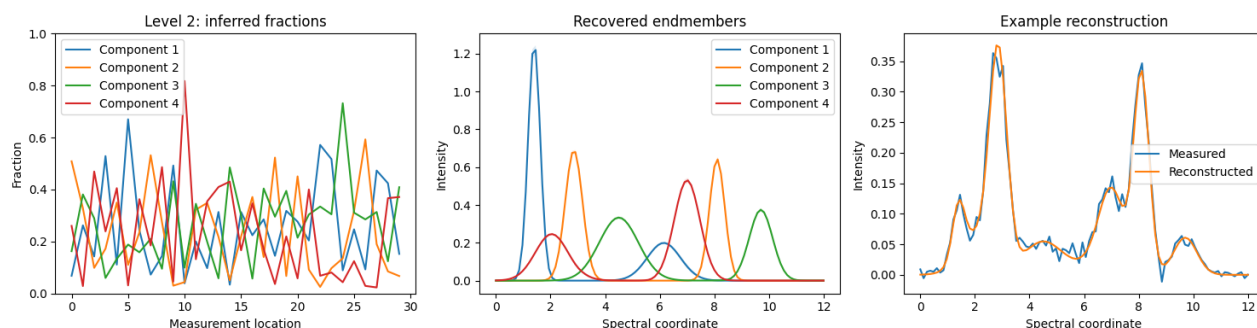


Figure 26. Notebook 10 execution. Level 2 result for experimental-style data: inferred fractions, recovered endmembers, and spectral reconstruction. Experimental interpretation must rely on posterior uncertainty, residuals, sensitivity, and independent validation rather than hidden truth.

12.1 Minimal experimental format

Quantity	Shape	Meaning
E	(M,)	Spectral coordinate shared by data and references.
Y	(N, M)	Measured spectra.
W_mean	(N, K)	Measured composition/fraction estimates.
W_sigma	(N, K)	Current public API: componentwise uncertainty scale in fraction space.
sigma_y	scalar or broadcastable to (N, M)	Measurement-noise SD.
S_fixed	(K, M), optional	Level 0 exact references.
theta_mean, theta_sigma	(K, P), optional	Level 1 parametric prior.
S_library	(K, R, M), optional	Level 2 ensemble of plausible complete spectra.

Simplex caution. The Integrated Framework treats composition uncertainty natively in ALR/ILR coordinates and emphasizes that arbitrary per-component SD tables may be incompatible with closure. The current v0.1 tutorial API accepts `W_sigma` for usability and uses a first-order ALR approximation internally. Near boundaries, with correlated assay errors, or for metrological work, a full simplex-native covariance model should replace this simplification.

13. Where should the uncertainty inputs come from?

The Integrated Framework's strongest practical message is that posterior calibration is only as credible as the provenance of the covariance objects. The package makes uncertainty explicit; it cannot determine automatically whether the supplied uncertainty describes the right random quantity.

13.1 Measurement noise

- Use detector physics, repeated analog measurements, blanks, or a documented preprocessing propagation model.
- If preprocessing aligns, smooths, normalizes, backgrounds-subtracts, or resamples channels, the resulting noise may be correlated; a diagonal `sigma_y` may then be inadequate.
- Residuals should be examined for energy-dependent structure and scale mismatch.

13.2 Composition uncertainty

- Use the independent assay's own error model, not the spread of compositions across the specimen.
- Respect closure: only $K-1$ directions are independent for a K -component composition.
- Carry correlated assay errors when elements share detector calibration, thickness corrections, or quantification models.
- Do not use covariance as a substitute for systematic assay bias.

13.3 Spectral covariance and Level 2 libraries

The source document identifies five routes: replicate pure-component measurements, parametric propagation, simulation ensembles, libraries/databases, and theory–experiment discrepancy calibration. These routes do not create the same object. A simulation ensemble, for example, represents uncertainty only after a distribution over physical inputs has been specified; a uniform parameter grid is an experimental design, not automatically a probability distribution.

Spectral-library provenance. Document whether the ensemble spread represents uncertainty about one shared endmember (the current Stage-1 model) or genuine sample-to-sample variability. Do not silently fold detector noise, instrument mismatch, and intrinsic material variability into one Level 2 library.

14. Validation checklist before trusting a posterior

1. Validate dimensions, finite values, positivity, and fraction closure.
2. Check spectral-grid and intensity conventions across data and references/libraries.
3. For a physical spectral covariance, check symmetry, positive semidefiniteness, and the unit-area null direction.
4. Run prior-predictive checks: can the supplied priors generate spectra resembling what the instrument actually observed?
5. Inspect standardized or appropriately normalized residuals for structure across spectral coordinate and location.
6. For synthetic studies, report both recovery error and interval coverage. Point estimates can remain good while uncertainty calibration fails.
7. Stress-test prior width, bias, functional rank, and measurement-noise assumptions separately.
8. For broad or strongly nonlinear priors, validate the Laplace approximation with a stronger posterior method before interpreting regime boundaries.

The Integrated Framework reports a recurring pattern across its larger studies: RMSE can be robust while coverage is fragile. Wrong noise scale, underestimated composition uncertainty, biased narrow spectral priors, or insufficient functional rank can leave point estimates near canonical while credible intervals become seriously miscalibrated. The tutorial notebooks teach the object structure; calibration requires repeated realizations or simulation-based validation.

15. A practical decision tree

Question	Use	Next notebook
Do I have trusted component reference spectra that I am willing to treat as exact?	Level 0	01
Do I have a physically meaningful low-dimensional line-shape model?	Level 1	02
Do I instead have replicates, simulations, or a curated ensemble of complete spectra?	Level 2	03, then 06–08
Are prior qualities very different across components?	Component-specific uncertainty vectors	04
Do I need to understand which information source controls the result?	Uncertainty trade-off sweep	05
Do I need spectral covariance/correlation rather than pointwise bands?	Level 2 covariance diagnostics	06
Do I need to understand the log-PCA representation itself?	Functional mode diagnostics	07
Do I need cross-component or concentration–spectral posterior dependence?	Joint posterior diagnostics	08
Am I moving to real measurements?	<code>from_arrays()</code> + <code>set_level*()</code>	10

16. What comes after Probabilistic Unmixing 0

The integrated source document explicitly separates the current Stage-1 problem from harder extensions. These should be added as new model layers rather than hidden inside an uncertainty scale:

- Local endmember variability: distinguish a global epistemic spectrum from location-specific intrinsic spectral perturbations.
- Spatial or temporal coupling: place priors on the appropriate latent object (fractions, labels, or local perturbations) without double-counting cross-location covariance already induced by shared global spectra.
- Composition-to-phase mapping: elemental assays and spectral phase fractions become distinct variables connected by stoichiometry and signal-yield physics.
- Solid solutions: endmember spectra become continuous functions of composition, often requiring a manifold or hierarchical phase-diagram model.
- Nonlinear mixing, thickness, absorption, plural scattering, and instrument response: place them in the forward model with their own nuisance parameters.
- Unknown or missing components and backgrounds: diagnose through structured residuals and solve by model comparison or sparse selection.
- Poisson counting and detector calibration: use count likelihoods and explicit expected-count models; do not represent counting statistics by an arbitrary Gaussian residual covariance.

17. Suggested exercises

Notebook	Exercise
00	Change N, M, and K-related inputs and verify that the data contract and plots remain consistent.
01	Increase one component's W uncertainty while holding S exact; compare posterior SD with recovery error.
02	Change spectral_uncertainty for one component and identify which spectral parameter family drives the largest deformation.
03	Change max_rank and compare spectral RMSE, posterior SD, and library reconstruction behavior.
04	Swap the good/poor assignments between Components 2 and 3 and check whether the posterior follows the information source or the component identity.
05	Expand to the full 3×3 or 5×5 grid and look for diagonal information-tradeoff boundaries.
06	Repeat covariance/correlation analysis for all four components and compare covariance scale with correlation structure.
07	Change the latent width by multiplying coefficients and observe how the physical mean moves under exponentiation/normalization.
08	Choose a different mixed location and different representative spectral channels; compare concentration–spectral coupling.
10	Replace the demonstration arrays with one real dataset and write down the provenance of every uncertainty input before fitting.

18. What to report in a real analysis

- Data dimensions, spectral coordinate units, normalization convention, and any preprocessing.
- How W_mean and its uncertainty were obtained, including simplex treatment and correlations.
- How sigma_y was estimated and whether channel correlations are ignored or modeled.
- Which spectral level was used and why.
- For Level 1: parameter definitions, prior means, prior covariance/SD, and bounds/transforms.
- For Level 2: library provenance, number of library members, preprocessing, latent representation, retained ranks, variance target, and held-out library reconstruction if available.
- Posterior fraction and spectral uncertainty, residual diagnostics, and predictive checks.
- Sensitivity to prior widths, bias, noise calibration, and Level 2 rank.
- Coverage or simulation-based calibration for synthetic/repeated-realization validation; do not rely on RMSE alone.
- Known omissions: local variability, background, instrument calibration, missing components, or nonlinear mixing.

19. API cheat sheet

```
# Synthetic tutorial data
data = pu.generate_example(
    concentration_uncertainty=[0.01, 0.01, 0.10, 0.10],
```

```

    spectral_uncertainty=[0.25, 2.0, 0.25, 2.0],
    seed=42,
)

# Experimental data
data = pu.from_arrays(E=E, Y=Y, W_mean=W_mean, W_sigma=W_sigma, sigma_y=sigma_y)

# Attach spectral information
data.set_level0(S_reference)
data.set_level1(model="two_gaussian", theta_mean=theta_mean, theta_sigma=theta_sigma)
data.set_level2(S_library)

# Fit
r0 = pu.fit(data, level=0)
r1 = pu.fit(data, level=1)
r2 = pu.fit(data, level=2)

# Inspect
pu.validate_data(data)
pu.print_data_summary(data)
pu.plot_data(data)
pu.plot_result(data, r2)
pu.plot_level_comparison(data, r0, r1, r2)
pu.plot_uncertainty_structure(data)

```

20. Selected background references from the Integrated Framework

1. Gemperline, P. J. & Cash, E. (2003). Advantages of soft versus hard constraints in self-modeling curve resolution problems. *Analytical Chemistry* 75, 4236–4243.
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4. Borsoi, R. A. et al. (2021). Spectral variability in hyperspectral data unmixing: a comprehensive review. *IEEE Geoscience and Remote Sensing Magazine* 9, 223–270.
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6. Phan, D.-T., Bobin, J. et al. (2024). Semi-blind spectral unmixing for gamma-ray spectrometry using an interpolating autoencoder. *NIM A* 1060, 169028.
7. Dobigeon, N. & Brun, N. (2012). Spectral mixture analysis of EELS spectrum-images. *Ultramicroscopy* 120, 25–34.
8. Teurtrie, A. et al. (2024). From STEM-EDXS data to phase separation and quantification using physics-guided NMF. *Machine Learning: Science and Technology* 5, 045050.
9. Kannan, R., Ziatdinov, M. & Kalinin, S. V. (2018). Physically constrained linear unmixing of multimodal hyperspectral data. *Advanced Structural and Chemical Imaging* 4, 6.
10. Aitchison, J. (1986). *The Statistical Analysis of Compositional Data*. Chapman and Hall.

Appendix A. Figure and execution provenance

Figures 1–26 were extracted from notebook outputs generated by executing the tutorial notebooks against a local reconstruction of probunmix v0.1.0. The execution used the notebook code and package interfaces developed for the Probabilistic-Unmixing-0 repository. Notebook 05 was executed on a reduced 2×2 grid for document generation; the distributed notebook defaults to a denser 3×3 sweep.

Figures	Notebook
1	00_Quick_Start_and_Data.ipynb
2	01_Level_0_Known_Endmembers.ipynb
3–4	02_Level_1_Parametric_Spectral_Uncertainty.ipynb
5–6	03_Level_2_Functional_Spectral_Uncertainty.ipynb
7–8	04_Heterogeneous_Component_Uncertainty.ipynb
9	05_Uncertainty_Tradeoff_Sweep.ipynb (reduced execution for tutorial document)
10–13	06_Level_2_Covariance_and_Correlation.ipynb
14–17	07_Level_2_Functional_Modes_and_Prior_Geometry.ipynb
18–24	08_Level_2_Posterior_Correlation_and_Propagation.ipynb
25–26	10_Experimental_Data_Template.ipynb